Design: Alternatives Analysis

CUNY EES 79903

Seminar on Nature-Based Solutions

Kyle McKay

November 3, 2026

Lecture Overview:

- 415-420 – Housekeeping
- 420-445 – Discussion with Amy Bredes
- 445-615 – Discussion: Hong Kong NBS Guide

	Leader #1	Leader #2
~~Sep 8: NBS fundamentals~~	~~Grace~~	~~Jesse~~
~~Sep 15: NBS fundamentals~~	~~Michelle~~	~~Shahela~~
~~Sep 22: Structured decision making and community engagement~~	~~Jesse~~	~~Aurelia~~
~~Sep 29: Analysis – ecological~~	~~Carol~~	~~Julia~~
~~Oct 6: Analysis – co-benefits~~	~~Shahela~~	~~Aurelia~~
~~Oct 20: Analysis – engr performance~~	~~Grace~~	~~Michelle~~
~~Oct 27: Analysis – engr performance~~	~~Abdellah~~	~~Julia~~
Nov 3: Alternatives analysis	Aurelia	Abdellah
Nov 10: Performance-based design	Aaryan	Michelle
Nov 17: Constructability	Grace	Aaryan
Nov 24: Performance monitoring	Shahela	Jesse
Dec 1: Performance monitoring	Julia	Carol
Dec 8: Life cycle analysis	Aaryan	Abdellah
Dec 15: Course synthesis	Kyle	Brett

Project Reminder

- ~~Oct 20 – Recommended timeframe to compile initial references~~
- ~~Nov 3 – Recommended timeframe for outlining ideas~~
- Nov 17 – Initial submission of traceable account
- Nov 24 – Review of traceable account by classmates
- Dec 8 – Final submission of traceable account
- Dec 15 – Course synthesis (McKay)

Topic	First option
Qualitative foundations of project planning	Jesse
Facilitation, engagement, and teaming	Aurelia
Ecological analysis of NBS	Julia
Economic analysis of NBS	
Co-benefits analysis of NBS	
Engineering analysis of NBS	Abdellah
Alternatives analysis and structured decision making	
Performance-based design	Aaryan
Constructability	Michelle
Performance monitoring	
Project establishment	
Asset management	Grace

Paper Discussion

Government of the Hong Kong Special Administrative Region. 2026. Hong Kong Nature-based Solutions Design Guidelines. Civil Engineering and Development Department and Agriculture, Fisheries, and Conservation Department.

NBS Principles

- Do these resonate with your expertise?
- What do they leave out?
- How do they help someone design?

Core Principle I:

Promoting Ecosystem Diversity at Multiple Scales

Guideline 1: Enhancing Habitat and Spatial Diversity at Macro Scale
Guideline 2: Diversifying Habitat Components at Intermediate Scale
Guideline 3: Creating Complex Micro-habitats

Core Principle II:

Embracing Human-Nature Coexistence for Mutual Benefits

Guideline 4: Addressing Societal Challenges through Nature-based Solutions
Guideline 5: Designing Human-Nature Composition in Context
Guideline 6: Harmonising Urban-Nature Transitions

Core Principle III:

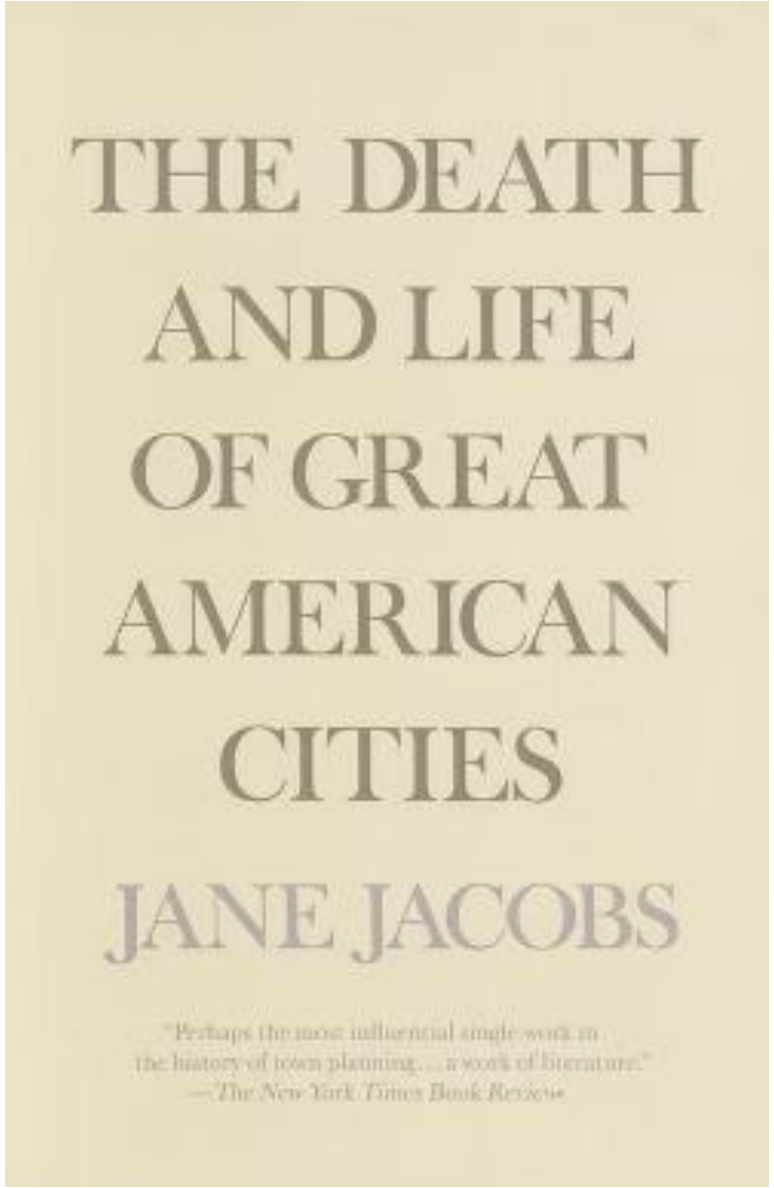
Improving Resilience through Nature-based Solutions

Guideline 7: Addressing Climate Challenges
Guideline 8: Building Ecological Resilience
Guideline 9: Promoting Socio-economic Sustainability

One of the unsuitable ideas behind projects is the very notion that they ***are*** projects, abstracted out of the ordinary city and set apart…The aim should be to get that project, that patch upon the city, rewoven back into the fabric – and in the process of doing so, strengthen the surrounding fabric too.

Jane Jacobs (1961)

Chapter 20 (emphasis original)



Next Week

Visitors:
Anna Hochalter and Nicole Del Monaco (Ramboll)

Papers:
Godsey, E.S., Everett, C.L., Tritinger, A.S., Bryant, D.B., Altman, S., Sowers, A.A., Bassetti, L., Bruck, J. and Bonura, C., 2026. Engineering practice guide for oyster reefs: a natural infrastructure approach for coastal systems.

Ostrow, K.S. and Cox, D.T., 2026. Performance-Based Design Methodology for Nature-Based Solutions: Using Mangroves to Mitigate Wave Overtopping. ASCE OPEN: Multidisciplinary Journal of Civil Engineering, 4(1), p.04026006.